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Aim

Implement and analyze various probability distributions in Python, including Binomial, Poisson, and Normal distributions.

✓ Theory:

Q1. Difference between Binomial, Poisson, and Normal distributions

Probability distributions describe how values of a random variable are distributed.

1. Binomial Distribution:

- It is a discrete probability distribution.
- It models the number of successes in a fixed number of independent trials.
- Each trial has only two outcomes: success or failure.
- Parameters:
 - $n \rightarrow$ number of trials
 - $p \rightarrow$ probability of success
- Example: Number of heads in 10 coin tosses.

2. Poisson Distribution:

- It is also a discrete distribution.
- It models the number of events occurring in a fixed interval of time or space.
- Events must occur independently.
- Parameter:
 - λ (lambda) $\rightarrow$ average number of events
- Example: Number of customers arriving in a store per hour.

3. Normal Distribution:

- It is a continuous distribution.
- It is symmetric and bell-shaped.
- Most values cluster around the mean.
- Parameters:
 - $\mu \rightarrow$ mean
 - $\sigma \rightarrow$ standard deviation
- Example: Heights of students in a class.

Key Differences:

- Binomial and Poisson are discrete, Normal is continuous.
- Binomial uses fixed trials, Poisson uses time intervals, Normal models natural data.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, poisson, norm

plt.figure(figsize=(15,5))

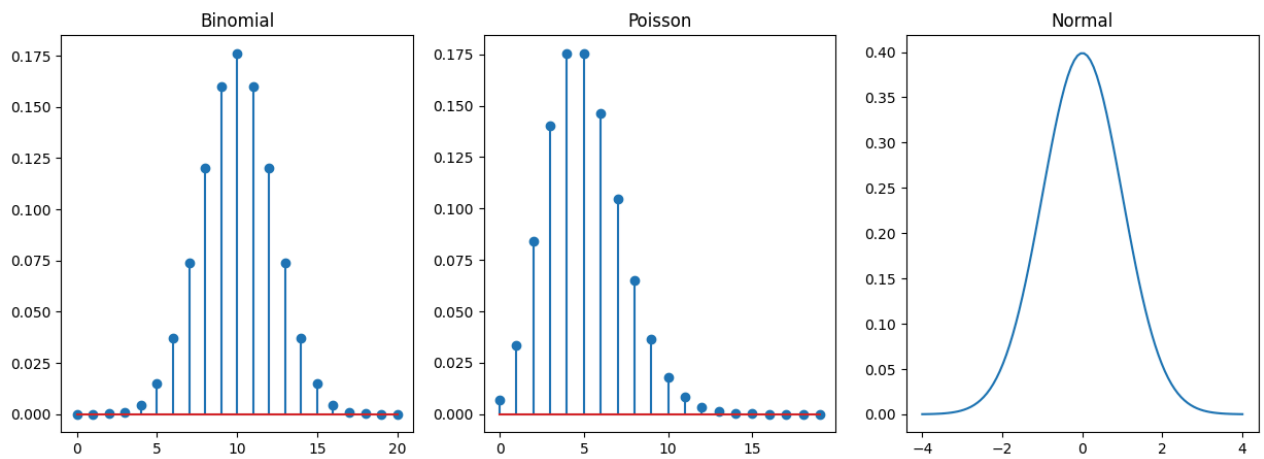
x = np.arange(0,21)
plt.subplot(1,3,1)
plt.stem(x, binom.pmf(x,20,0.5))
plt.title('Binomial')

x = np.arange(0,20)
plt.subplot(1,3,2)
plt.stem(x, poisson.pmf(x,5))
plt.title('Poisson')

x = np.linspace(-4,4,100)
plt.subplot(1,3,3)
plt.plot(x, norm.pdf(x,0,1))
```

```
plt.title('Normal')
```

```
plt.show()
```



✓ Q2. Real-world significance of Poisson distribution

Theory:

The Poisson distribution is widely used in real-world applications where events occur randomly over time.

Conditions:

- Events occur independently
- The average rate remains constant
- Two events cannot occur at exactly the same instant

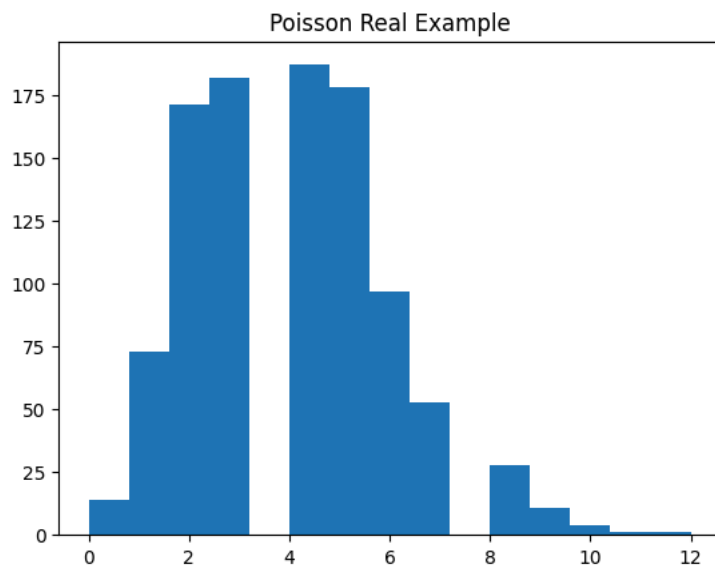
Applications:

- Number of calls received in a call center
- Number of accidents at a traffic junction
- Number of defects in manufacturing
- Number of emails received per hour

This distribution helps in predicting probabilities of rare events.

```
import numpy as np
import matplotlib.pyplot as plt

data = np.random.poisson(lam=4, size=1000)
plt.hist(data, bins=15)
plt.title('Poisson Real Example')
plt.show()
```



✓ Q3. Central Limit Theorem (CLT)

Theory:

The Central Limit Theorem is one of the most important concepts in statistics.

It states that:

- When we take repeated samples from any distribution,
- The distribution of the sample means approaches a normal distribution
- This happens even if the original data is not normally distributed

Conditions:

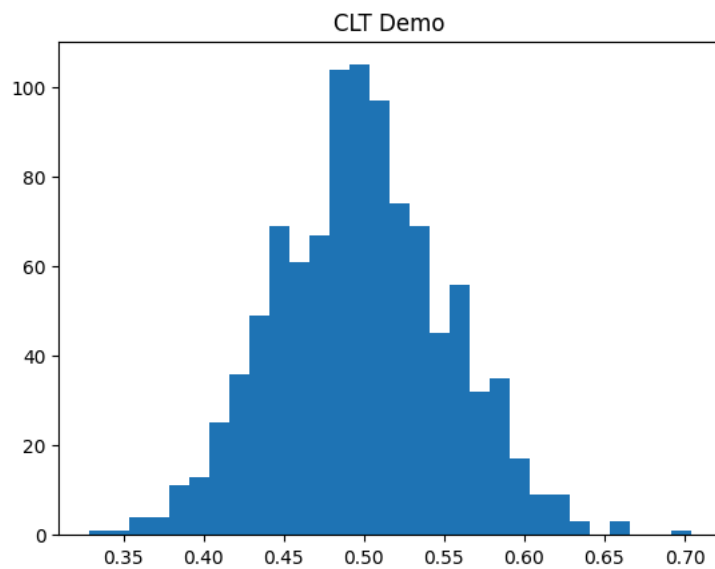
- Sample size should be sufficiently large (usually $n \geq 30$)
- Samples should be independent

Importance:

- Forms the basis of statistical inference
- Used in hypothesis testing and confidence intervals

```
import numpy as np
import matplotlib.pyplot as plt

data = np.random.uniform(0,1,10000)
means=[]
for i in range(1000):
    sample=np.random.choice(data,30)
    means.append(np.mean(sample))
plt.hist(means, bins=30)
plt.title('CLT Demo')
plt.show()
```



Q4. Why Normal distribution is widely used

Theory:

The Normal distribution is widely used because many real-world variables follow it.

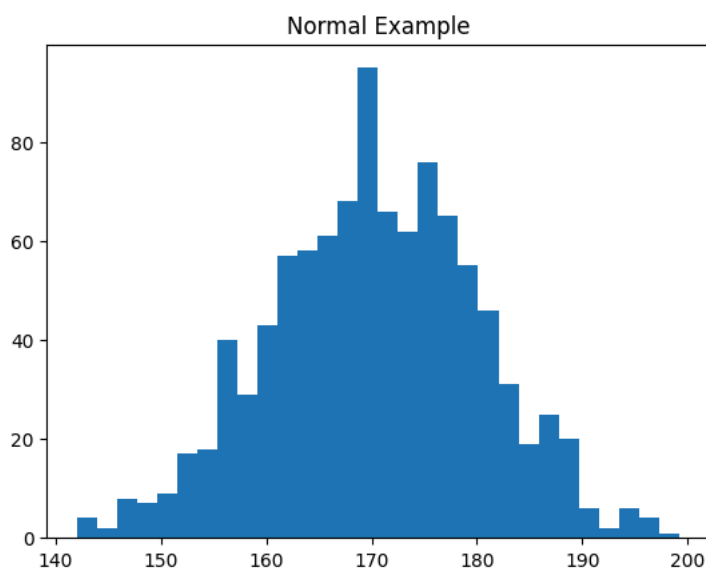
Reasons:

- Many natural phenomena are normally distributed (heights, weights, marks)
- It simplifies mathematical calculations
- Many statistical methods assume normality

In Machine Learning:

- Used in algorithms like Naive Bayes
- Used in error modeling
- Helps in feature scaling and analysis

```
import numpy as np
import matplotlib.pyplot as plt
heights=np.random.normal(170,10,1000)
plt.hist(heights,bins=30)
plt.title('Normal Example')
plt.show()
```



Q5. Skewness in distributions

Theory:

Skewness measures how much a distribution deviates from symmetry.

Types:

- Positive skew: Long tail on right side
- Negative skew: Long tail on left side
- Zero skew: Perfectly symmetric distribution

Observations:

- Normal distribution has zero skewness
- Binomial and Poisson distributions can be skewed depending on parameters

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import skewnorm
pos=skewnorm.rvs(a=10,size=1000)
neg=skewnorm.rvs(a=-10,size=1000)
plt.subplot(1,2,1)
plt.hist(pos,bins=30)
plt.title('Positive Skew')
plt.subplot(1,2,2)
plt.hist(neg,bins=30)
plt.title('Negative Skew')
plt.show()
```

